Exp No:10

Experiment to understand array function in Data science.

Program:

import numpy as np

array=np.random.randint(1,100,9)

Output:

```
import numpy as np
array=np.random.randint(1,100,9)
array

array([13, 37, 42, 20, 75, 89, 84,  8, 71])
```

np.sqrt(array)

array.ndim

Output:

```
np.sqrt(array)

array([3.60555128, 6.08276253, 6.4807407 , 4.47213595, 8.66025404,
       9.43398113, 9.16515139, 2.82842712, 8.42614977])

array.ndim

1
```

new_array=array.reshape(3,3)

new_array

new_array.ravel()

Output:

```
new_array=array.reshape(3,3)
new_array

array([[13, 37, 42],
       [20, 75, 89],
       [84,  8, 71]])

new_array.ravel()

array([13, 37, 42, 20, 75, 89, 84,  8, 71])
```

nw_array.ndim

newm=new_array.reshape(3,3)

newm

Output:

```
new_array.ndim

2

newm=new_array.reshape(3,3)
newm

array([[13, 37, 42],
       [20, 75, 89],
       [84,  8, 71]])
```

newm[2,1:3]

newm[1:2,1:3]

new_array[0:3,0:0]

new_array[0:2,0:1]

new_array[0:3,0:1]

new_array[1:3]

Output:

```
newm[2,1:3]

array([ 8, 71])

newm[1:2,1:3]

array([[75, 89]])

new_array[0:3,0:0]

array([], shape=(3, 0), dtype=int64)

new_array[0:2,0:1]

array([[13],
       [20]])
```

```
new_array[0:3,0:1]

array([[13],
       [20],
       [84]])

new_array[1:3]

array([[20, 75, 89],
       [84,  8, 71]])
```